

KERNEL DENSITY ESTIMATION (KDE)-BASED DEFENCE AGAINST MODEL POISONING IN FEDERATED LEARNING

PRESENTED BY:

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ABSTRACT

Research Context

Federated Learning (FL) enables collaborative model training across decentralised devices whilst preserving data privacy. However, this distributed architecture introduces vulnerabilities to model poisoning attacks.

Proposed Solution

We introduce a KDE-based defence mechanism that detects and filters malicious model updates by analysing their density distribution within the parameter space.

Key Contribution

The KDE approach provides a statistical foundation for identifying anomalous updates without requiring complete knowledge of client data distributions or extensive computational overhead.

OBJECTIVES

01

Investigate Vulnerabilities

Analyse security weaknesses in standard federated averaging algorithms against model poisoning attacks

02

Develop KDE Framework

Design a kernel density estimation approach for detecting anomalous model parameter updates

03

Implement Filtering Mechanism

Create adaptive threshold strategies for rejecting malicious contributions whilst preserving legitimate updates

04

Evaluate Performance

Validate defence effectiveness across multiple datasets and attack scenarios

EXISTING SYSTEM: FEDERATED LEARNING ARCHITECTURE

Central Server

Aggregates model updates using FedAvg algorithm and broadcasts global model

Client Selection

Randomly selects subset of clients for each training round

Local Training

Clients train on local data and submit model updates

Model Update

Global model improves iteratively through aggregation



CRITICAL DRAWBACKS OF CURRENT SYSTEMS

No Built-in Security

Standard FedAvg assumes honest participants and lacks mechanisms to verify update authenticity or detect malicious behaviour.

Single Point of Failure

A compromised server or malicious client can poison the global model, affecting all participants without detection.

Computational Overhead

Existing defence mechanisms like differential privacy or cryptographic verification add significant computational and communication costs.

False Positives

Statistical methods struggle to distinguish between legitimate model variance and malicious updates, potentially rejecting valid contributions.

LITERATURE SURVEY

Title	Author(s)	Technology / Methodology	Advantages	Drawbacks
LoMar: A Local Defense Against Poisoning Attack on Federated Learning (2022)	X. Li, Z. Qu, et al.	**Relative KDE** scoring based on client neighborhoods.	Effective against label-flipping; lightweight; high accuracy retention.	Performance may degrade in extremely high-dimensional weight spaces.
Beta Poisoning Attacks: Extensions, Limitations and Defenses (2022)	A. Kara, N. Koprucu, et al.	**Gaussian/Epanechnikov KDE** + GAN Discriminator.	Comprehensive evaluation of KDE kernels; GAN-based detection of noisy samples.	Defense is primarily focused on linear models; less effective against complex CNNs.
KAFÈ: Kernel Aggregation for FEderated Learning (2024)	P. Qi, D. Chiaro, et al.	**KDE-constructed** global classification layers.	Addresses non-IID data issues; reduces client drift; low computational overhead.	Requires specific layer-wise weight distribution estimation.
FedRDF: A Robust and Dynamic Aggregation Function (2024)	E. M. Campos, et al.	**Fourier Transform** + Frequency-domain Density.	No prior knowledge of attacker count needed; handles sophisticated collusive attacks.	Increased mathematical complexity; relies on frequency domain stability.
Adversarial Defense in Cybersecurity: Systematic Review (2025)	Systematic Survey	**Statistical Density Analysis** + GAN-based filtering.	Provides a unified taxonomy for density-based threat detection.	High-level survey; lacks specific implementation parameters for FL.
KeTS: Kernel-based Trust Segmentation (2025)	A. Gangwal, M. Conti, et al.	**Historical Update KDE** for trust segmentation.	Outperforms FLTrust by 24% on MNIST; robust against benign outliers.	Higher memory usage due to tracking client update evolution history.
FLAMED: Federated Learning via Aggregating Densities (2025)	W. Briguglio	**Multivariate KDE** for global model simulation.	Bypasses non-IID drift; ensures high data privacy by never aggregating raw weights.	Scaling multivariate density estimation to millions of parameters is difficult.
A Defense Method Against Multi-Label Poisoning Attacks (2025)	W. Ma, Q. Zhao, et al.	**Gap Statistic** + K-means + Cluster Compactness (Density).	Defends against multi-target attacks; identifies benign clusters with high precision.	Dependent on the quality of output layer gradient features.
TDF-PAD: Two-step Defense Framework for Poisoning (2025)	IEEE Access (PP)	**Interquartile Range (IQR)** + Z-score Density Filtering.	Achieves 0% false positive rate on benchmark datasets; very low false rejection.	Step-wise approach may increase latency in massive-scale FL rounds.
Securing FL: A Defense Strategy Against Targeted Attacks (2025)	Discover IoT (2025)	**PCA** + Anomaly Density Detection.	Effectively isolates malicious updates across vulnerable training rounds.	PCA dimensionality reduction may discard subtle but important model features.

PROPOSED SYSTEMS

Software Requirements

- Python 3.8+ with NumPy and SciPy libraries
- PyTorch or TensorFlow for FL implementation
- Scikit-learn for KDE computation
- OpenSSL for secure communications
- Matplotlib for visualisation

Hardware Requirements

- Central server: 16+ GB RAM, 4+ cores CPU
- Client devices: Minimum 4 GB RAM
- Network bandwidth: 10 Mbps upload/download
- Storage: 50+ GB for model checkpoints

200+

Client Devices

Scalable to hundreds of participants

<5%

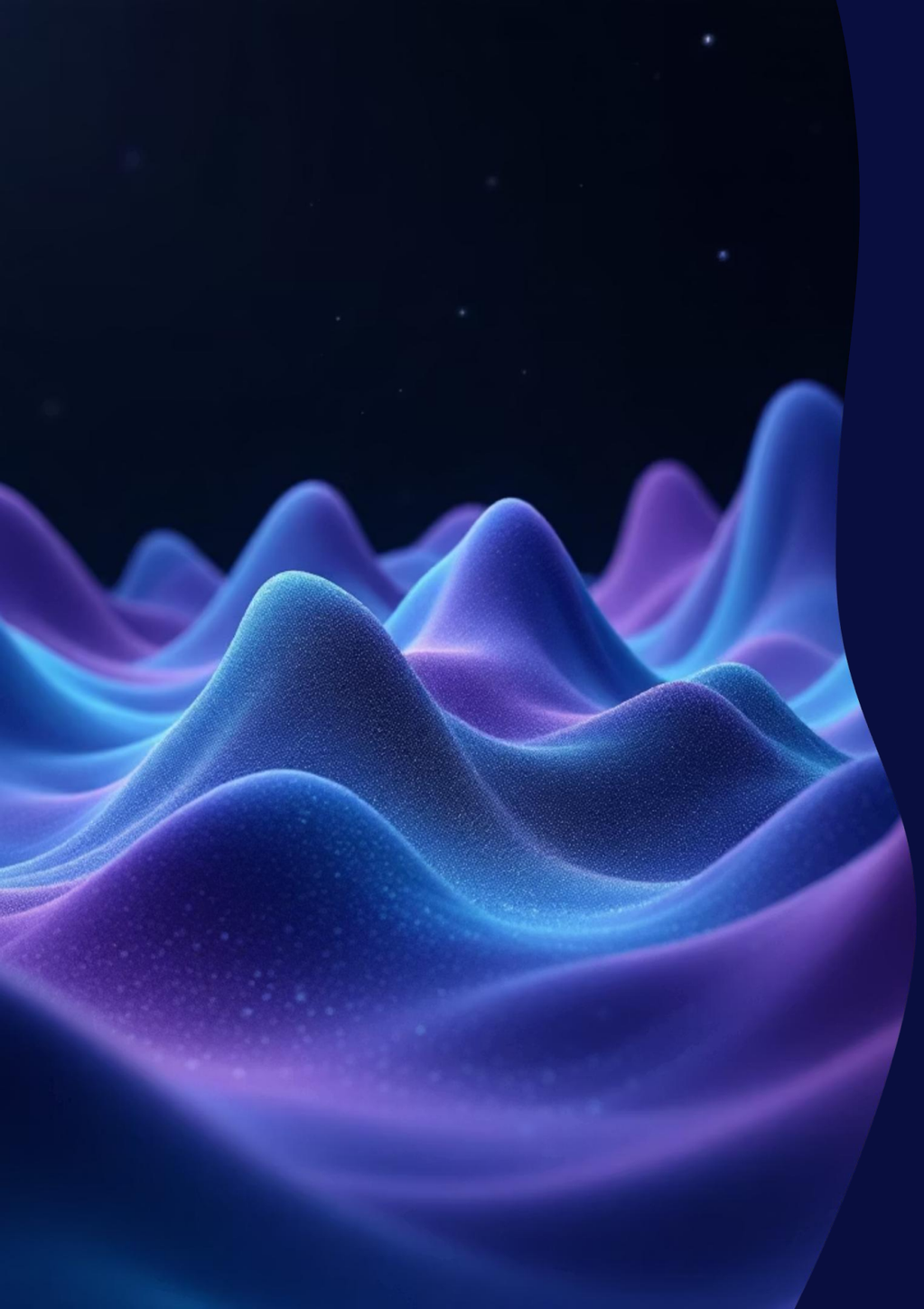
False Positive Rate

Target threshold for legitimate updates

95%+

Detection Accuracy

Expected attack identification rate



PROPOSED KDE-BASED DEFENCE SYSTEM

Key Advantages

Statistical Robustness

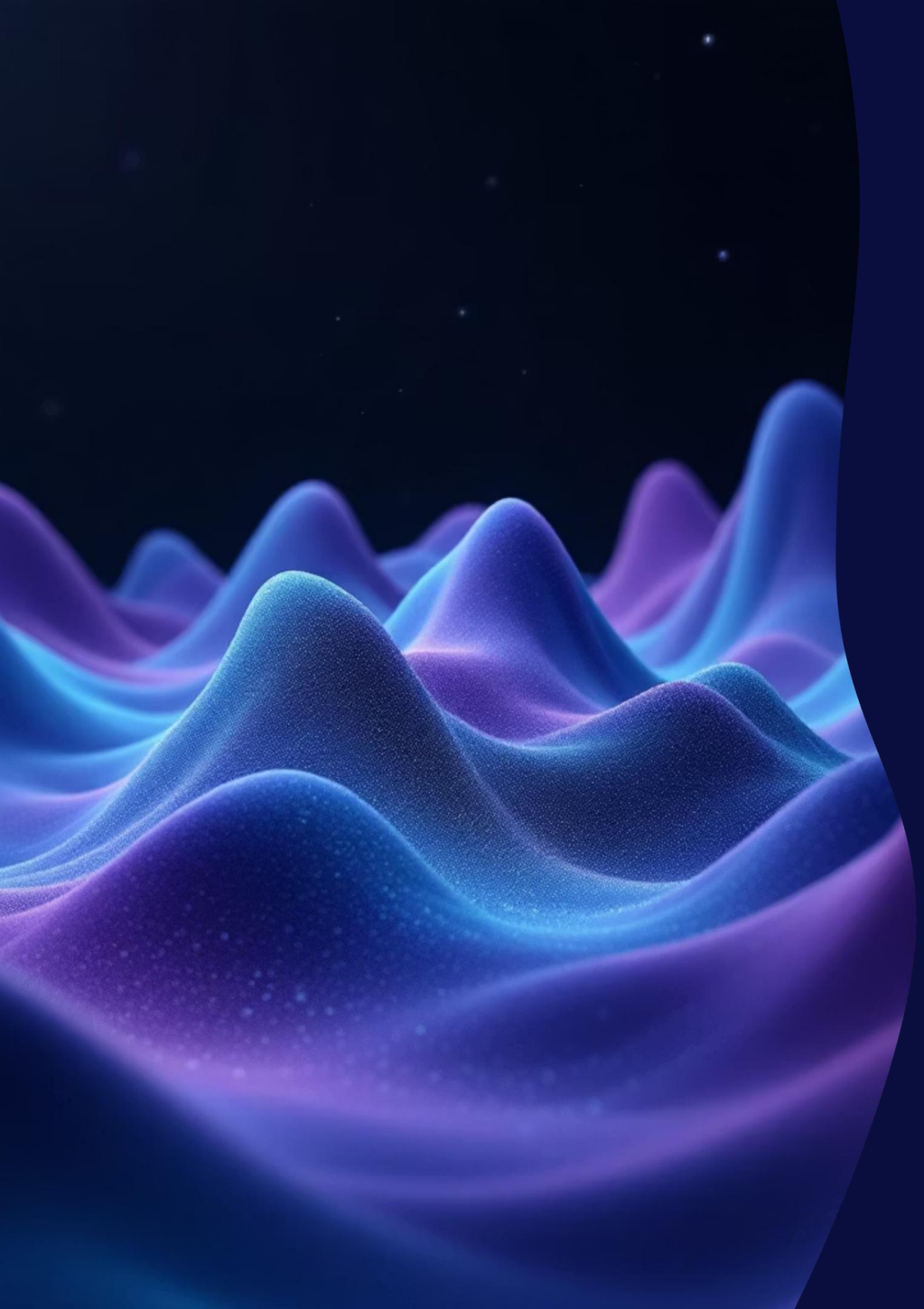
KDE provides non-parametric density estimation that adapts to complex parameter distributions without distributional assumptions.

Low Overhead

Computational cost scales efficiently with model size, adding minimal latency to the aggregation process.

Adaptive Thresholding

Dynamic threshold calculation based on density scores reduces false positives whilst maintaining attack detection.



MODULES

1. Dimensionality Reduction Module
Transforms high-dimensional model gradients into a lower-dimensional latent space to overcome the "curse of dimensionality" during density estimation.

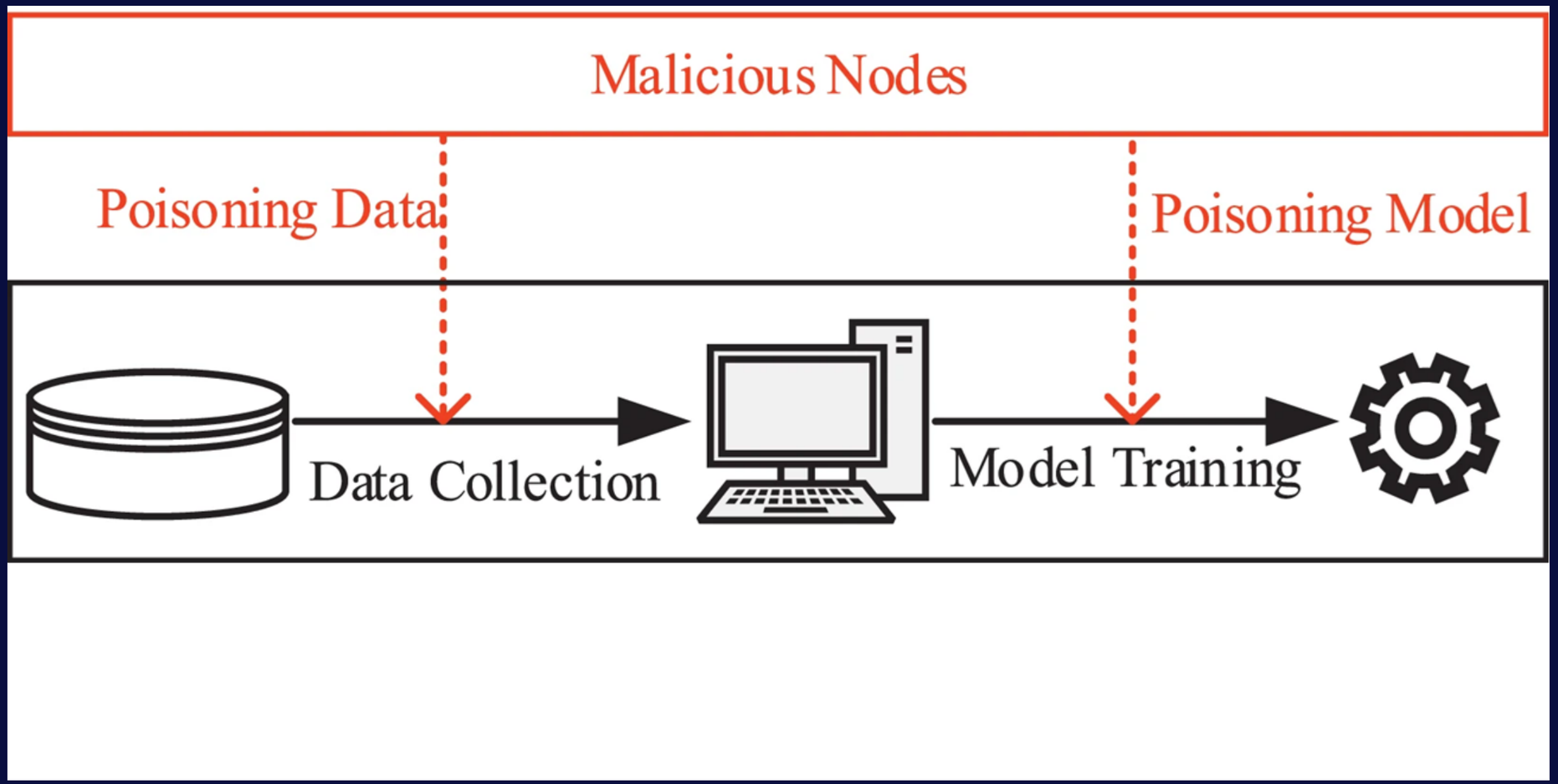
2. Kernel Selection & Bandwidth Tuning Module
Selects the appropriate mathematical function (e.g., Gaussian) and smoothing parameter to accurately model the distribution of benign client updates.

3. Density Scoring & Probability Mapping Module
Calculates a numerical likelihood score for each client's update based on its position within the estimated global probability distribution.

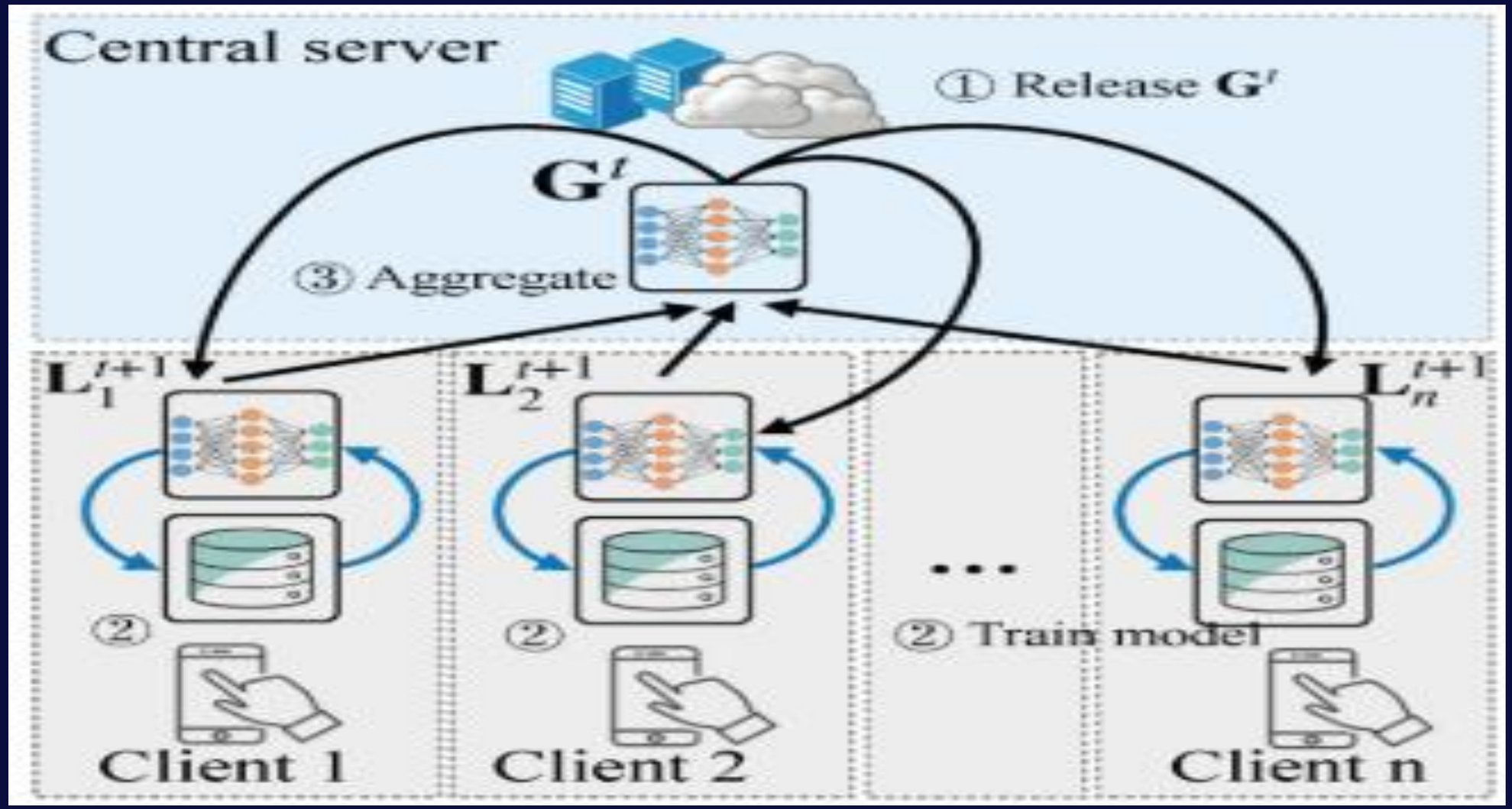
4. Statistical Outlier Detection Module
Identifies and flags malicious model updates that fall into low-density regions, signaling they are statistical anomalies compared to the honest majority.

5. Robust Aggregator Module
Combines the filtered "clean" updates into a new global model, often weighting them by their density scores to prioritize the most representative contributions.

SYSTEM ARCHITECTURE



DATAFLOW DIAGRAM



ALGORITHM USED

1. Bandwidth Selection Algorithms

•Silverman's Rule of Thumb: Computes a smoothing parameter based on the data's standard deviation to provide a quick, Gaussian-based density baseline.
•Sheather-Jones (SJ) Algorithm: Solves a fixed-point equation to find the mathematically optimal bandwidth that minimizes error for non-normal, multi-modal distributions.
•Cross-Validation (CV): Uses an iterative "leave-one-out" approach to select the bandwidth that best fits the actual observed client update patterns.

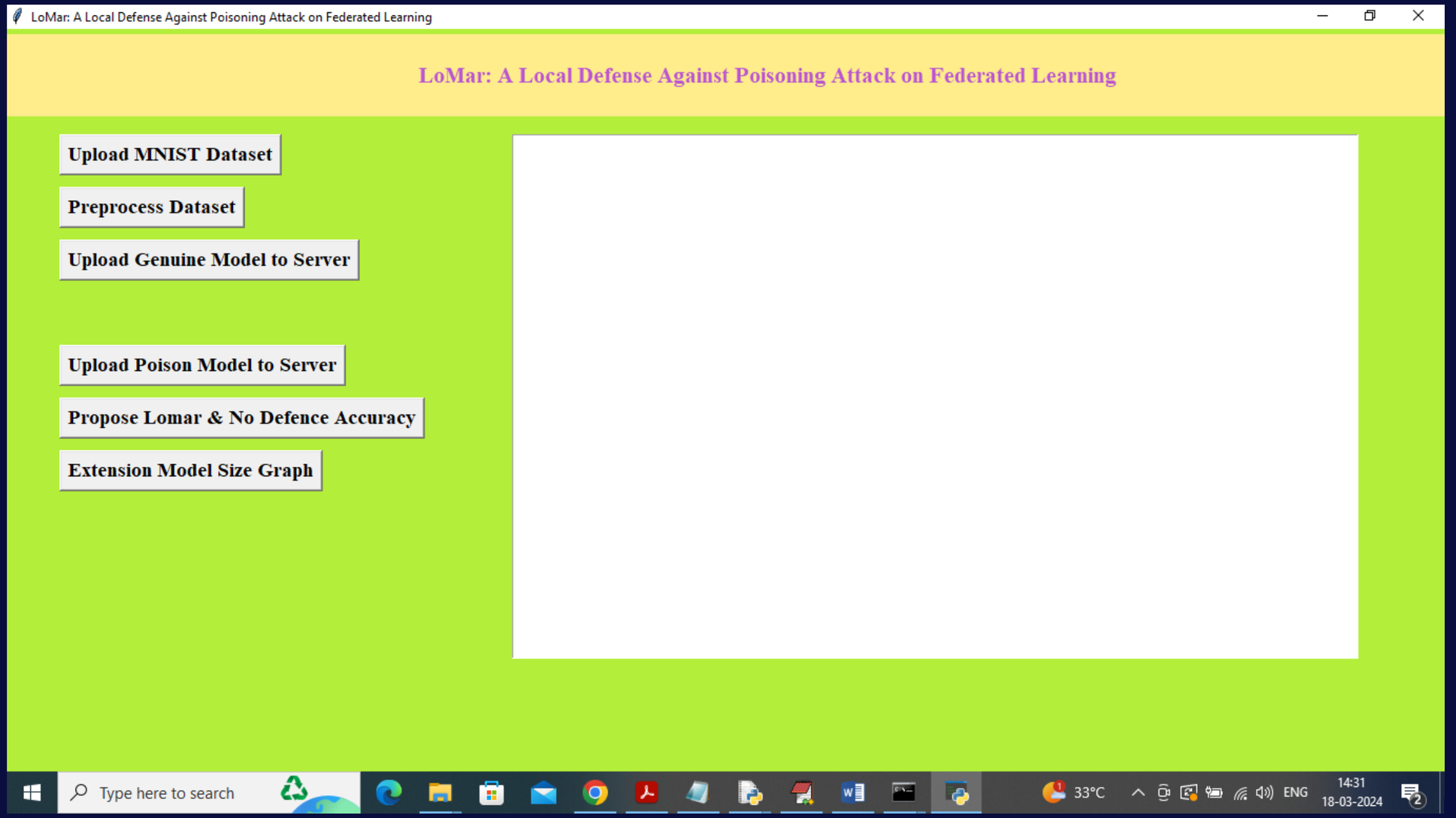
2. Fast Computation & Approximation Algorithms

•Fast Multipole Method (FMM): Accelerates density estimation by grouping distant client updates into spatial clusters and approximating their collective influence using Taylor series.
•Dual-Tree Recursion: Uses hierarchical data structures like KD-Trees to prune entire groups of distant updates, focusing only on immediate statistical neighbors.
•Linear Binning & FFT: Maps updates onto a discrete grid and uses Fast Fourier Transforms to compute density convolutions in near-constant time.

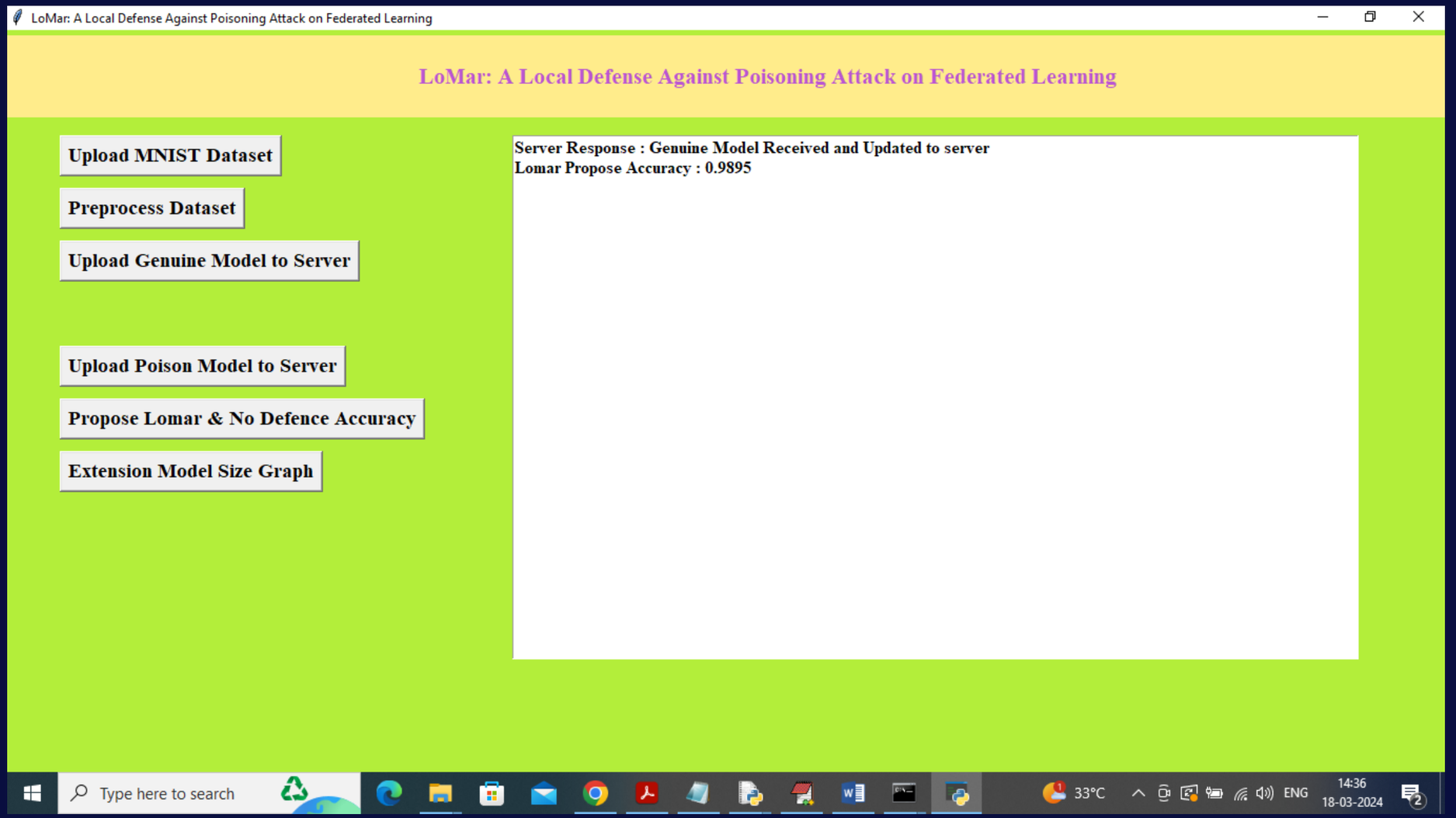
3. Outlier Scoring & Detection Algorithms

•Relative Density (LoMar): Compares a client's density score to its local neighborhood average to prevent unique but honest (non-IID) updates from being flagged.
•Kernel Density Ratio (KDR): Measures the statistical divergence between the current round's updates and a pre-verified "golden" dataset to identify malicious shifts.

USER VIEW CODE



OUTPUT


LoMar: A Local Defense Against Poisoning Attack on Federated Learning
LoMar: A Local Defense Against Poisoning Attack on Federated Learning
Upload MNIST Dataset
Preprocess Dataset
Upload Genuine Model to Server
Upload Poison Model to Server
Propose Lomar & No Defence Accuracy
Extension Model Size Graph
Server Response : Genuine Model Received and Updated to server
Lomar Propose Accuracy : 0.9895
Type here to search
33°C
ENG
14:36
18-03-2024

CONCLUSION

Effective Defence

KDE-based filtering significantly reduces model poisoning success rates whilst maintaining model accuracy on legitimate data.

Practical Implementation

Minimal computational overhead makes the approach deployable in real-world federated learning systems.

Future Directions

Integration with differential privacy and adaptive threshold optimisation for enhanced security guarantees.

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Questions & Discussion

"Security in federated learning requires continuous innovation to stay ahead of evolving attack vectors."

